Računanje reda algoritma za problem maksimalne sume podniza

Za zadani algoritam je potrebno izračunati red izvršavanja ukoliko su osnovne operacije:

- računanje trenutne sume,
- usporedba trenutne sume s maksimalnom.

Primjer 1:

```
        int MaxSumaPodniza1(int A[], int N) {
/*1*/           int trenSuma, maxSuma = 0, i, j, k;
/*2*/           for(i = 0; i < N; i++)
/*3*/                   for(j = i; j < N; j++)
                      {
/*4*/                           trenSuma = 0;
/*5*/                           for(k = i; k<=j; k++)
/*6*/                                   trenSuma += A[k];
/*7*/                           if(trenSuma > maxSuma)
/*8*/                                   maxSuma = trenSuma;
                      }
/*9*/           return maxSuma;
        }
```

$$T(n) = \sum_{i=0}^{N-1}\sum_{j=i}^{N-1}\left(\sum_{k=i}^{j} 1 + 1\right)$$

$$= \sum_{i=0}^{N-1}\sum_{j=i}^{N-1}(j+1-i+1) = \sum_{i=0}^{N-1}\left(\sum_{j=i}^{N-1} j + \sum_{j=i}^{N-1} 2 - \sum_{j=i}^{N-1} i\right)$$

$$= \sum_{i=0}^{N-1}\left(\sum_{j=1}^{N-1} j - \sum_{j=1}^{i-1} j + 2\sum_{j=i}^{N-1} 1 - i\sum_{j=i}^{N-1} 1\right)$$

$$= \sum_{i=0}^{N-1}\left(\frac{(N-1)N}{2} - \frac{(i-1)i}{2} + 2(N-1+1-i) - i\,(N-1+1-i)\right)$$

$$= \frac{1}{2}\sum_{i=0}^{N-1}(N^2 - N - i^2 + i + 4N - 4i - 2iN + 2i^2)$$

$$= \frac{1}{2}\left[(N^2+3N)\sum_{i=0}^{N-1} 1 - (3+2N)\sum_{i=0}^{N-1} i + \sum_{i=0}^{N-1} i^2\right]$$

$$= \frac{1}{2}\left[(N^2+3N)N - (3+2N)\frac{(N-1)N}{2} + \frac{(N-1)N(2N-2+1)}{6}\right]$$

$$= \frac{N^3+6N^2+5N}{3}$$

Red izvršavanja algoritma je: $\theta(N^3)$

Primjer 2:

```
        int MaxSumaPodniza2(int A[], int N)
        {
                int trenSuma, maxSuma, i, j;
                maxSuma = 0;
/*1*/           for(i = 0; i < N; i++)
                {
/*2*/                   trenSuma = 0;
/*3*/                   for(j = i; j < N; j++)
                        {
/*4*/                           trenSuma += A[j];
/*5*/                           if(trenSuma > maxSuma)
/*6*/                                   maxSuma = trenSuma;
                        }
                }
                return maxSuma;
        }
```

$$T(n) = \sum_{i=0}^{N-1}\sum_{j=i}^{N-1}(1+1) = 2\sum_{i=0}^{N-1}\sum_{j=i}^{N-1}1 = 2\sum_{i=0}^{N-1}(N-i) = 2N(N) - 2\frac{(N-1)N}{2} = 2N^2 - N^2 + N$$
$$= N^2 + N$$

Red izvršavanja algoritma je: $\theta(N^2)$

Primjer 4:

```
        int MaxSumaPodniza4(int A[], int N)
        {
                int maxSum, trenSum, i;
/* 1*/          maxSum = trenSum = 0;
/* 2*/          for(i = 0; i<N; i++)
                {
/* 3*/                  trenSum += A[i];
/* 4*/                  if(trenSum > maxSum) maxSum = trenSum;
/* 5*/                  else if(trenSum < 0) trenSum = 0;
                }
/* 6*/          return maxSum;
        }
```

$$T(n) = \sum_{i=0}^{N-1}(1+1) = 2\sum_{i=0}^{N-1}1 = 2(N-1+1) = 2N$$

Red izvršavanja algoritma je: $\theta(N)$